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COMET FWC 071

Seventh International Olympiad, 1965

1965/1

Determine all values x in the interval $0 \leq x \leq 2\pi$ which satisfy the inequality

$$2\cos x \leq \left|\sqrt{1+\sin 2x} - \sqrt{1-\sin 2x}\right| \leq \sqrt{2}.$$

1965/2

Consider the system of equations

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = 0$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = 0$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = 0$$

with unknowns x_1, x_2, x_3. The coefficients satisfy the conditions:

(a) a_{11}, a_{22}, a_{33} are positive numbers.

(b) The remaining coefficients are negative numbers.

(c) In each equation, the sum of the coefficients is positive.

Prove that the system has only the trivial solution $x_1 = x_2 = x_3 = 0$.

1965/3

Given a tetrahedron $ABCD$ whose edges AB and CD have lengths a and b respectively. The distance between the skew lines AB and CD is d, and the angle between them is ω. The tetrahedron is divided by a plane ε, parallel to lines AB and CD. The ratio of distances of ε from AB and CD is k. Compute the ratio of volumes of the two solids obtained.

1965/4

Find all sets of four real numbers x_1, x_2, x_3, x_4 such that the sum of any one and the product of the other three is equal to 2.

1965/5

Consider $\triangle OAB$ with acute angle AOB. Through a point M, perpendiculars are drawn to OA and OB, with feet P and Q respectively. The intersection of the altitudes of $\triangle OPQ$ is H. Find the locus of H when M varies over:

1. the side AB
2. the interior of $\triangle OAB$

1965/6

In a plane a set of n points ($n > 3$) is given. Each pair of points is connected by a segment. Let d be the length of the longest of these segments. We define a diameter of the set to be any connecting segment of length d. Prove that the number of diameters of the given set is at most n.